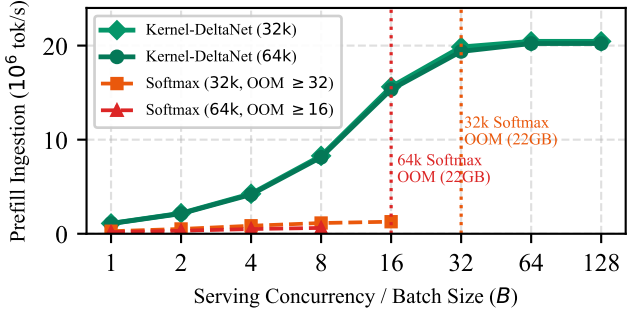
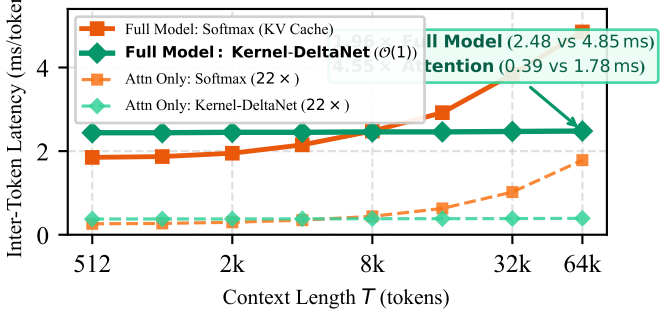

(a) Prefill Ingestion Scaling
Kernel-DeltaNet (32k)
Kernel-DeltaNet (64k)
Softmax (32k, OOM ≥ 32)
Softmax (64k, OOM ≥ 16)
32k Softmax
OOM (22GB)
64k Softmax
OOM (22GB)
Prefill Ingestion (10^6 tok/s)
0
10
20
1
2
4
8
16
32
64
128
Serving Concurrency / Batch Size (B)


(b) Full-Model vs Attention Decoding ITL ($B = 1$)
Full Model: Softmax (KV Cache)
Full Model : Kernel-DeltaNet ($\mathcal{O}(1)$)
Attn Only: Softmax (22 ×)
Attn Only: Kernel-DeltaNet (22 ×)
ll Model (2.48 vs 4.85 ms)
tention (0.39 vs 1.78 ms)
Inter-Token Latency (ms/token)
0
2
4
512
2k
8k
32k
64k
Context Length T (tokens)